

Project Design Phase

Solution Architecture

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Project Name	ShopEZ
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Solution Architecture:

The ShopEZ Solution Architecture defines the overall structure of the online shopping platform and illustrates how the frontend, backend, and database interact to provide a secure, scalable, and efficient e-commerce application. It bridges the gap between customer requirements and technology by organizing the system into different layers that communicate through RESTful APIs.

The architecture follows the MERN Stack and the MVC design pattern. The React frontend provides an interactive user interface for customers and administrators. The Node.js and Express.js backend exposes REST APIs that handle authentication, product management, cart operations, order processing, and user management. MongoDB stores all application data, including users, products, carts, categories, and orders.

Example - Solution Architecture Diagram:

